

Eigenvalues and inverse problems

The smallest-multiplier rule for nearest unit-absolute-determinant matrices

Difficulty: challenging **Importance:** interesting to specialist**Status:** [Lean verified](#)

Rating rationale: Challenging reflects a global root-selection rule across all dimensions and singular-value data; specialist impact is the exact projection onto a specific determinant constraint.

Resolution — 2026-09-11

Negative resolution by Matthew J. Colbrook (Department of Applied Mathematics and Theoretical Physics, University of Cambridge). See the [complete manuscript](#), **Theorem SP-04, sections 1–4** ([PDF](#) · [LaTeX](#)), prepared 11 September 2026.

The stationary pair with uniquely smallest absolute multiplier fails to minimize the Frobenius distance on a nonempty open set of real 3×3 data matrices with distinct singular values in $(7/4, 44/25)$. Both determinant signs are allowed, and the open-set argument refutes the algebraic-generic formulation.

The original proof draft was generated in a ChatGPT conversation. A separate Codex agent independently verified the full proof and its match to the exact target on 11 September 2026: [detailed PASS review](#). The review records a hash of the unchanged proof text. This is independent agent verification, not external human peer review or formal certification. [The submission history and diagnostic record](#) preserve the initial solution claim. The ratings above are historical, and the earlier literature checks below are retained.

Lean verification — 15 September 2026

The complete original question has a formally verified negative answer: on a nonempty open subset of the full space of real three-by-three matrices, the unique least absolute stationary multiplier selects a feasible matrix that is not nearest. This family meets the complement of every proper algebraic exception, refuting the generic rule. All nine reviewed statements passed kernel-only LeanCert trust audits, standard-axiom checks, and [fresh isolated Linux Comparator/default-kernel verification](#) at the [immutable proof revision](#).

Formalization: Sidney Holden, with OpenAI Codex assistance. Matthew J. Colbrook retains mathematical authorship. Two independent statement reviews preceded implementation, and two independent final code reviews passed. Actual rejection and sandbox controls passed. Reviews are AI-agent reviews, not external human peer review or official Tau Ceti endorsement.

[Proof and reviews](#) · [Retained Linux evidence](#).

Original problem statement

For $n \geq 2$, define

$$\mathcal{G}_n = \{X \in \mathbb{R}^{n \times n} : \det X \in \{-1, 1\}\}, \quad \|Y\|_F^2 = \sum_{i,j} y_{ij}^2.$$

For a real data matrix U , consider the real stationary pairs

$$\mathcal{C}(U) = \{(X, c) \in \mathcal{G}_n \times \mathbb{R} : X^T(U - X) = cI_n\}.$$

For generic U , this set is finite. Choose a pair (X_*, c_*) whose multiplier has minimum absolute value. Does

$$\|U - X_*\|_F = \min_{X \in \mathcal{G}_n} \|U - X\|_F$$

hold for every $n \geq 2$ and generic real U ?

Here generic means outside an exceptional proper real algebraic set; in particular U is invertible and its squared singular values are distinct. If minimum absolute multipliers tie, the question can be restricted to the generic locus where the choice is unique. The source phrases the same selection question using the real roots of its elimination polynomial $R_1(c)$, which correspond one-to-one to stationary matrices for generic data.

Relevance

This is a concrete question about selecting the global nearest matrix from the stationary solutions of a structured matrix nearness problem. A positive answer would supply a scalar criterion for identifying the global solution. Both determinant signs are allowed; this must not be silently replaced by projection onto $\text{SL}_n(\mathbb{R})$ alone.

References

- J. A. Baaijens and J. Draisma, *Euclidean distance degrees of real algebraic groups*, Linear Algebra and its Applications 467 (2015), 174–187, §4.1, equations (4)–(5), elimination algorithm, and Problem 4.3 on p.186 ([version of record](#); [journal](#)).
- P. Jaap and O. Sander, *How to project onto $SL(n)$* , arXiv:2501.19310 (2025), introduction pp.1–2, discussing the Baaijens–Draisma conjectural global selection rule ([primary manuscript](#)); Journal of Optimization Theory and Applications 209 (2026), article 40 ([journal](#)).

Status check — 2026-09-08

The January 2025 manuscript explicitly describes the earlier proposed global-minimum selection as conjectural. Its journal version appeared in April 2026; the accessible journal abstract and arXiv version history were checked, but the subscription-only journal body was not inspected. Searches for “nearest determinant-one matrix”, “Baaijens”, “smallest absolute value”, “projection $SL(n)$ ”, and 2025–2026 found no resolution of this exact rule. The problem retains the source’s generic-data scope.

Audit update — 2026-09-10

Rechecked [Baaijens–Draisma](#), §4.1, Problem 4.3, and [Sander’s 2025 manuscript](#), introduction. The latter still describes the global selection step as conjectural. Searches for smallest-multiplier projection results found no general proof. The positive-determinant projection problem and algorithms reaching stationary points must be distinguished from the stated $\det = \pm 1$ global rule; impact is narrowed to specialist.